

# Template Practices Document

## Contents

|   |                          |   |
|---|--------------------------|---|
| 1 | Worked Solution Examples | 1 |
| 2 | Graph Command Examples   | 3 |

## Evaluated criteria

C1 Example criterion description.

C2 Example criterion description.

C3 Example criterion description.

## 1 Worked Solution Examples

---

### Problems:

---

1.1. Example Uniform Investment Plan (C1 C2)

1.2. Example Linear Density Model (C2)

1.3. Example Table and Probability Summary (C3)

---

### 1.1 Example Uniform Investment Plan (C1 C2)

A variable  $V$  follows a uniform model on  $[8, 16]$ . Use the graph and density model to answer the tasks.

#### Questions/Tasks



C1 Write the density function and support.

C2 Compute  $P(10 \leq V \leq 14)$  and justify with area.

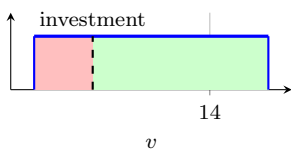
#### C1 - Write the density function and support

$$f_V(v) = \begin{cases} \frac{1}{16-8} = \frac{1}{8}, & 8 \leq v \leq 16, \\ 0, & \text{otherwise.} \end{cases}$$

$$8 \leq v \leq 16 \quad \longrightarrow \quad f_V(v) = \frac{1}{8} \quad \longrightarrow \quad \int_8^{16} f_V(v) dv = 1$$



**C2 - Compute  $P(10 \leq V \leq 14)$  and justify with area.**



$$P(10 \leq V \leq 14) = \int_{10}^{14} \frac{1}{8} dv = \frac{1}{8}(14 - 10) = \frac{4}{8} = 0.5$$

The shaded area represents half of the support interval.



## 1.2 Example Linear Density Model (C2)

Suppose  $X$  has density  $f(x) = kx$  for  $0 \leq x \leq 2$ .

**Questions/Tasks**

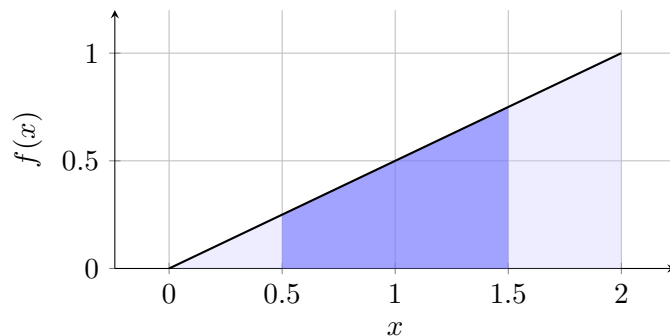


**C2.1** Find the value of  $k$ .

**C2.2** Compute  $P(0.5 \leq X \leq 1.5)$ .

**C2.1 - Find the value of  $k$ .**

$$1 = \int_0^2 kx dx = k \left[ \frac{x^2}{2} \right]_0^2 = 2k \Rightarrow k = \frac{1}{2}$$



**C2.2 - Compute  $P(0.5 \leq X \leq 1.5)$ .**

$$P(0.5 \leq X \leq 1.5) = \int_{0.5}^{1.5} \frac{1}{2}x dx = \left[ \frac{x^2}{4} \right]_{0.5}^{1.5} = \frac{1.5^2 - 0.5^2}{4} = 0.5$$



## 1.3 Example Table and Probability Summary (C3)

A summary table is shown for two intervals of a piecewise density.

**Questions/Tasks**

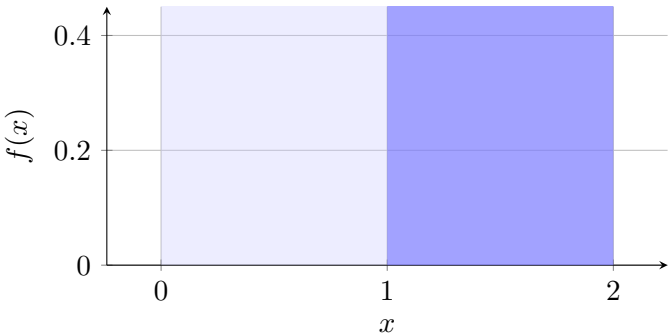


C3.1 Complete the probability summary table.

C3.2 State one consistency check.

C3.1 - Complete the probability summary table.

| Interval | Expression           | Result |
|----------|----------------------|--------|
| $[0, 1]$ | $\int_0^1 0.4 \, dx$ | 0.4    |
| $[1, 2]$ | $\int_1^2 0.6 \, dx$ | 0.6    |
| Total    | $0.4 + 0.6$          | 1      |



C3.2 - State one consistency check.

A valid check is that all interval probabilities are nonnegative and add to 1.



2 Graph Command Examples

This section provides one representative command from each graph file in book/preamble/graphs.

|                                                                             |
|-----------------------------------------------------------------------------|
| <b>Problems:</b>                                                            |
| 2.1. Example from <code>uniform_linear_probability_distributions.tex</code> |
| 2.2. Example from <code>probability_distribution_graphs.tex</code>          |
| 2.3. Example from <code>normal_distribution_graphs.tex</code>               |

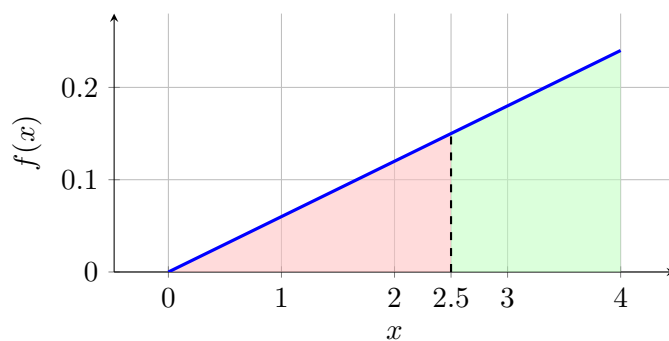
2.1 Example from `uniform_linear_probability_distributions.tex`.

Questions/Tasks



C2 Plot the linear density and shade the region below the threshold.

C2 - Plot the linear density and shade the region below the threshold.



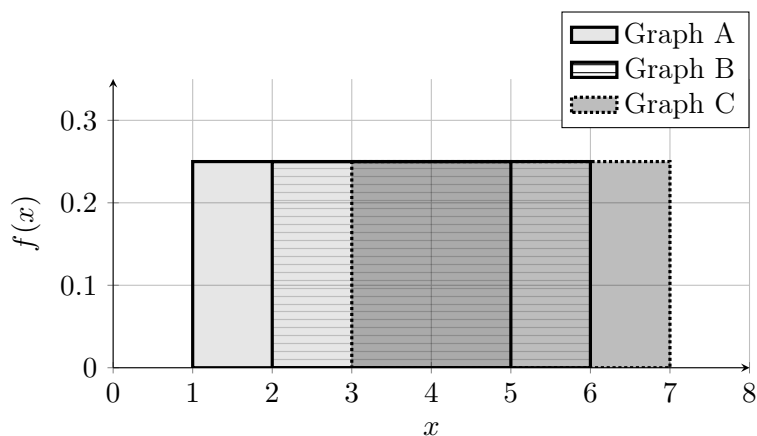
2.2 Example from probability\_distribution\_graphs.tex.

Questions/Tasks



C2 Plot the uniform density with three shaded intervals.

C2 - Plot the uniform density with three shaded intervals.



2.3 Example from normal\_distribution\_graphs.tex.

Questions/Tasks



C2 Plot the standard normal density and shade  $P(Z \leq -0.75)$ .

C2 - Plot the standard normal density and shade  $P(Z \leq -0.75)$ .

